Fragrance Sales Performance Analysis

1. Project Overview

This project provides a comprehensive analysis of a global fragrance dataset (2,000+ records) spanning markets like India, Canada, and United States. The goal was to transform raw, messy retail data into an interactive decision-making tool that tracks revenue trends, inventory health, and consumer demographic preferences.

2.Project Objective

The main objectives of this project are:

- To transform messy fragrance sales data into a clear, interactive dashboard that tracks revenue and stock levels.
- By connecting different data tables, the project allows users to instantly identify top-selling products and shopping trends across different countries and genders.

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2. Data Sources

- **Source:** Web scrapping
- **Domain**: Sales

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3. Problem Statement

- **To analyse revenue distribution** across different countries and fragrance types to identify high-performing markets and top-selling product categories.
- **To evaluate consumer behaviour** by segmenting sales data by gender and product type to understand demographic-specific purchasing patterns.
- **To examine the relationship** between price points, sales volume, and inventory levels to identify stock risks and pricing efficiency.
- **To study sales and update trends** over time to monitor market activity and improve operational timing and financial planning.
- **To develop an interactive Power BI dashboard** that transforms messy, siloed fragrance data into clear, actionable business insights through a structured data model.

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4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Type	Text	Type of perfume
Price	Currency	Rate of perfume
Available	Number	Stock quantity
Sold	Number	Sold quantity

Total Revenue	Currency	Total sales amount
Last updated date	Short Date	Date of sales
Gender	Text	Customer Gender
Country	Text	Country where customers belong to

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5. Tools & Technologies

- **Excel:** Data cleaning and transformation.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

6. Data Pre-Processing (Excel)

Tasks Performed:

- **Data Cleaning & Transformation:**
- Standardized formats
- Created calculated field

  1.Total_Revenue = =[@PRICE]*[@sold]
- Applied clean and Trim formulas to Type column.
- Separated Last updated Date and time column into Date, time and time zone using delimiter.
- Extracted country name from Item location column using formula

  Country =TRIM(RIGHT(SUBSTITUTE(A2, ",", REPT(" ", LEN(A2))), LEN(A2)))
- **Sorting:** Sorted date column from Newest to oldest.
- Convert the data into Fact and Dimension Table**.** Created 2 dimension tables.
  1. Type_Dim
  2. Country_Dim
  3. Extracted Type ID and Country ID from dimension tables to fact tables using Xlookup function.

     E.g.,

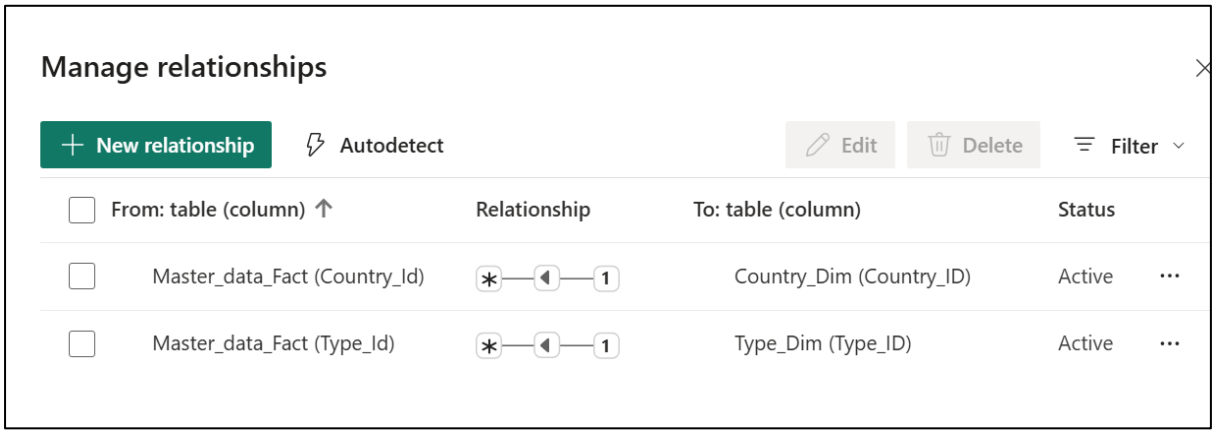
     =XLOOKUP(A2, Type_Dim!B:B, Type_Dim!A:A)

  Likewise Country_ID column.

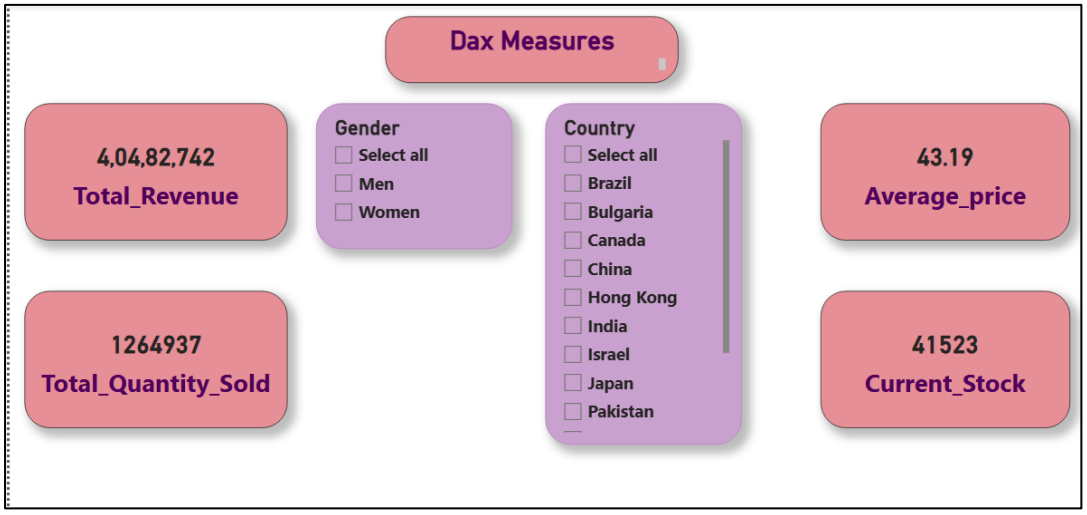
7. Data Modelling and DAX (Power BI)

- **Data Model:** Established relationships between tables, defined cardinality, and created lookup tables where necessary.

SCREENSHOT OF DATA MODELLING



- **DAX Measures:** Implemented DAX formulas for key metrics, such as
  1. Total_Revenue = sum(Master_data_Fact[Total_Revenue])
  2. Total_Quantity_Sold = Sum(Master_data_Fact[sold])
  3. Average_price = Average(Master_data_Fact[PRICE])
  4. Current_Stock = sum(Master_data_Fact[Available])



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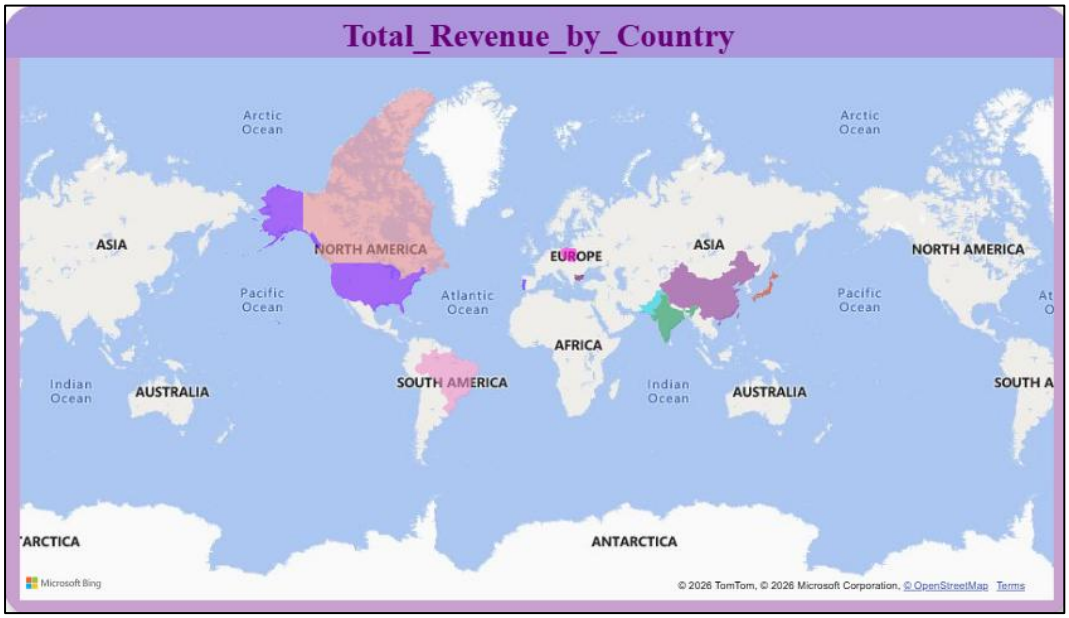
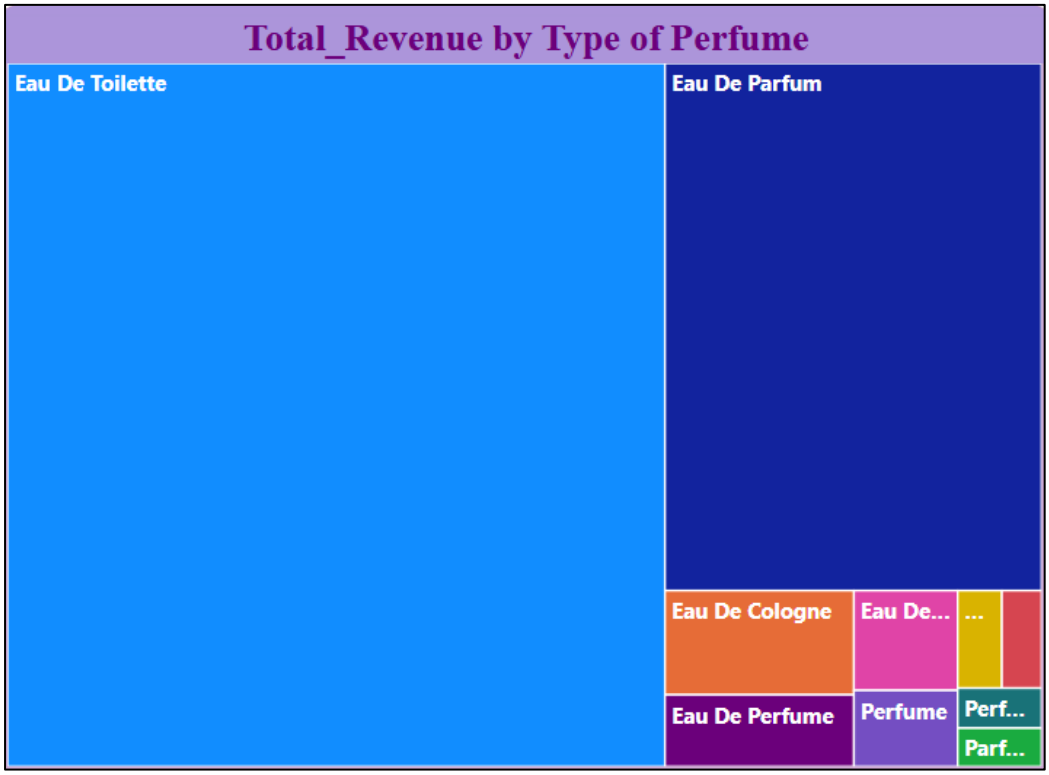
8. Analysis and Visualizations (Power BI)

Dashboard Features:

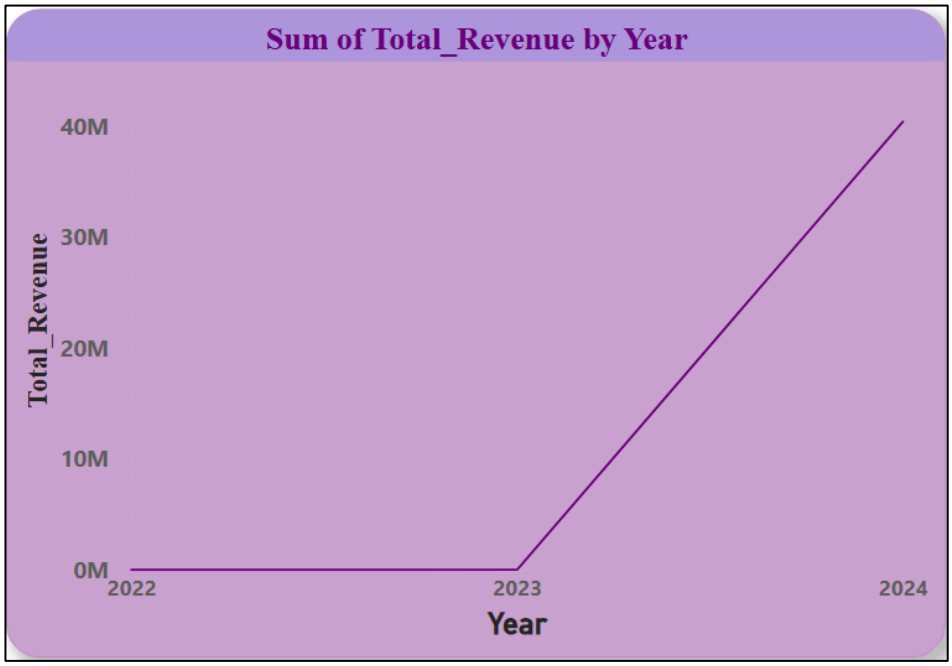
- **Multiple Visualizations based on problem statement:**

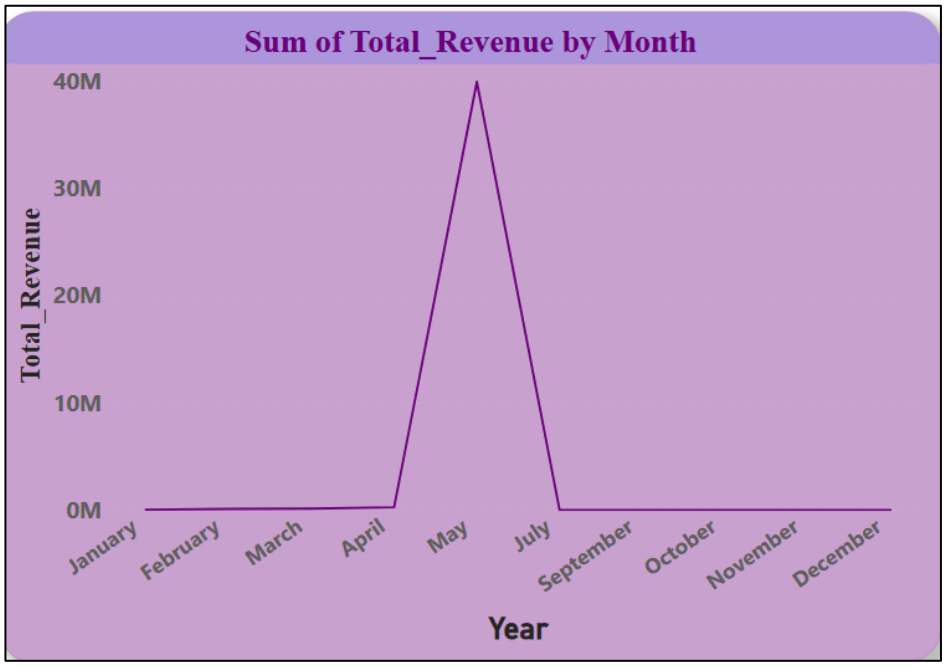


- Map visualization helps understanding the geographical view of Fragrance sales and total revenue distribution across the country.

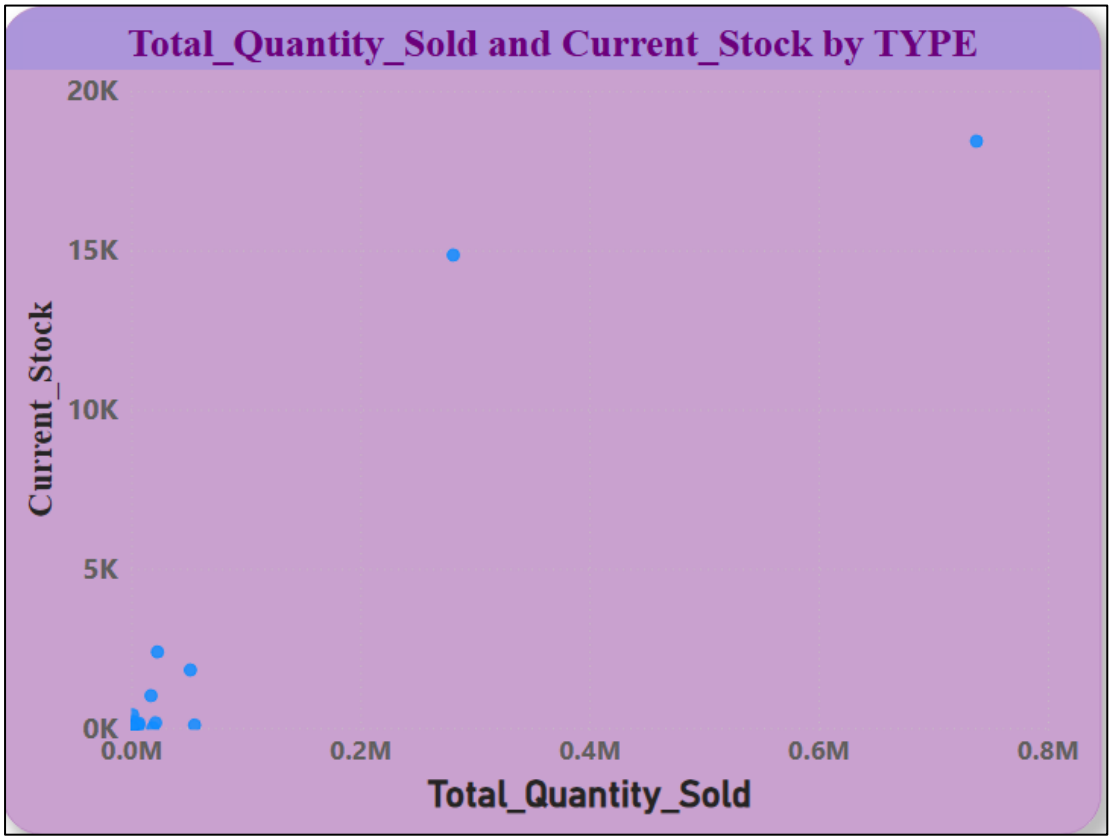


- Filtered top 10 types of perfume by their total revenue.
- Eau De Toilette ranks number 1 and Eau De Parfum ranks 2.





- This chart plotted to identify the trend of perfume sales across years, quarters and months.
- In the year 2024 only a greater number of perfumes sold and in the month of May.



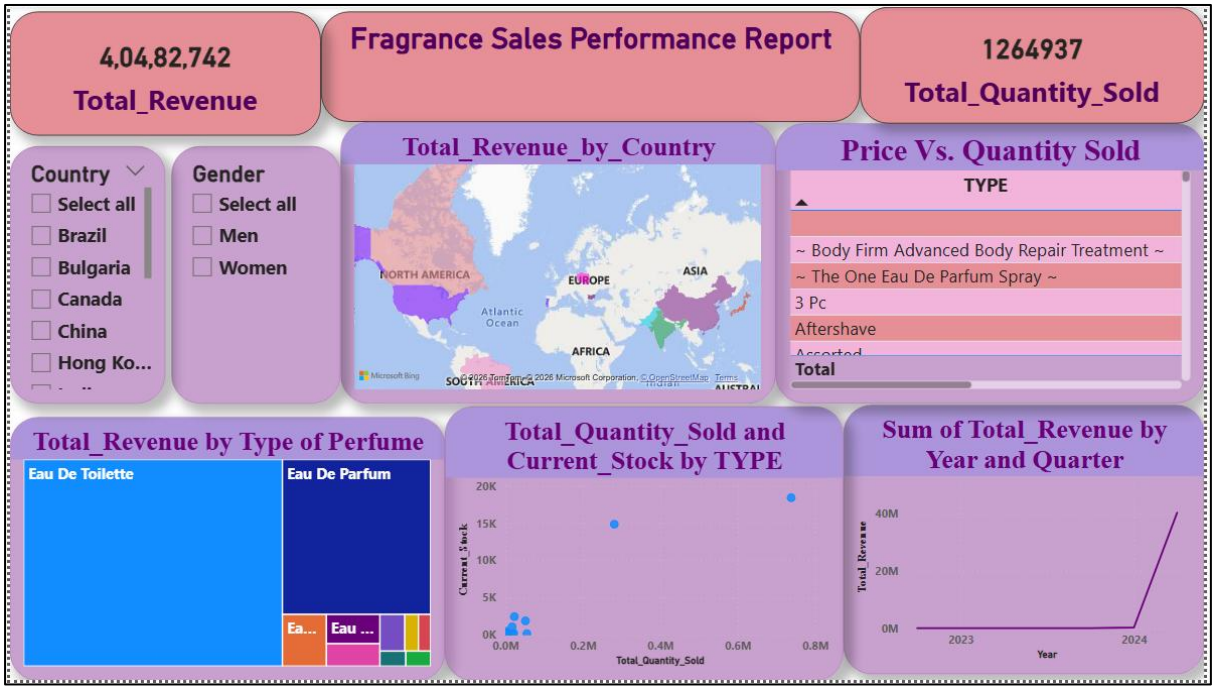
- **Bottom-Right Quadrant:** High Sales + Low Stock. These are our "Best Sellers" that are about to run out. We need to reorder these **immediately**.
- **Top-Left Quadrant:** Low Sales + High Stock. These are "Slow Movers." We might want to suggest a discount or a **"Buy 1 Get 1"** deal for these.

Price Vs. Quantity Sold

TYPE	Average_price	Total_Quantity_Sold	Total_Revenue
Various	20.00	2	40.00
Unscented	39.95	353	14,102.35
Toilette Spray	22.48	30	674.40
Spray	49.39	232	6,715.86
Splash-On	19.97	15	299.55
Solid Perfume Stick	5.65	178	1,005.70
Skin_Moisturizer	21.90	135	3,032.07
Shimmering Body Oil	34.99	30	1,049.70
Shimmer	25.00	79	1,975.00
Sensuous Body Moisturizer	28.40	65	1,846.00
Scented Oils	12.99	49	636.51
Roll On	9.99	94	939.06
Pink Sugar	33.50	11	368.50
Pheromone	12.49	21	261.79
Perfume, Eau De Parfum	51.73	6216	1,75,693.75
Perfume Oil	18.00	520	6,360.19
Perfume Gift Sets	18.87	1	18.87
Perfume Fragrance Mist	21.95	85	1,865.75
Perfume	36.10	17253	4,12,886.55
Total	**43.19**	**1264937**	**4,04,82,742.35**

- It allows us to see the exact numbers while providing a visual "bar" that makes comparisons instant. It’s perfect for spotting which "Types" have a high price but low volume versus those that have a low price but high volume.

REPORT:



9. Insights

- **Analysis insights:**
  - ❖ Based on the **Revenue by Country** map, **United States** and **Hong Kong** emerged as the highest revenue-generating markets, suggesting these regions have the highest brand penetration.
  - ❖ Markets like **Pakistan** showed steady growth but lower total volume, identifying them as prime candidates for targeted marketing campaigns.
  - ❖ The **Treemap** and **Revenue by Type** visuals clearly show that **Eau De Toilette** and **Eau De Parfum** account for over **60%** of total revenue.
  - ❖ Specialty types like **Parfum** and **Perfume Eau De Parfum** have high profit margins but lower sales frequency, indicating a "boutique" buyer profile.
  - ❖ Through the **Gender Slicer**, we discovered that the **Women's** segment drives significantly higher volume in "Parfum" categories, while the **Men's** segment prefers "Toilette" and "Cologne" types.
  - ❖ The **Price vs. Quantity sold** analysis identified several high-demand items (High Sold Quantity) that are currently running on **Low Available Stock**, presenting a risk of lost revenue.
  - ❖ Certain high-priced items in the **Bottom-Left** of our scatter plot are not moving as quickly as expected, suggesting a need for seasonal discounts or promotional bundling.
  - ❖ There are no Women customers in Brazil, Poland, Israel and Portugal. Whereas there are no Men customers in Bulgaria, Japan and Pakistan.

- Can start a targeted digital marketing campaign (social media ads) specifically for the "missing" gender in those countries.
- Using **Price vs. Popularity** table can find our globally most popular product for that gender and test it in the new market at a discount.

10.Conclusion:

The integration of Excel and Power BI proved effective for end-to-end data analysis, from raw data to visual reporting.